# HW3 - Numerical Analysis

Tanner Crane

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## 1 Suppose that $x > 0$ and $y < 0$ Using definition of absolute value and splitting into cases, show that $|x + y| \leq |x - y|$.

Definition of absolute value:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Consider $|x - y|$: Because $x > 0$ and $y < 0$, it follows that $-y > 0$. Therefore, $x - y$ is the sum of two positive numbers, so

$$|x - y| = x - y$$

Next, evaluate the left-hand side, $|x + y|$. Since $x$ is positive and $y$ is negative, the sum $x + y$ can be positive, negative, or zero.

Case 1: $x + y \geq 0$

If $x + y \geq 0$, the definition of absolute value states that $|x + y| = x + y$. Substituting into the inequality, we get

$$x + y \leq x - y \implies y \leq -y \implies 2y \leq 0$$

Since we are given that $y < 0$, the inequality $2y \leq 0$ is true. Therefore, $|x + y| \leq |x - y|$ holds in this case.

Case 2: $x + y < 0$
If $x + y < 0$, the definition of absolute value states that $|x + y| = -(x + y) = -x - y$. Substituting into the inequality, we get

$$-x - y \leq x - y \implies -x \leq x \implies 0 \leq 2x$$

Since we are given that $x > 0$, the inequality $0 \leq 2x$ is true. Therefore, $|x+y| \leq |x - y|$ holds in this case as well. Because the inequality holds true under all possible conditions for $x+y$, it is proven that $|x+y| \leq |x-y|$ for all $x > 0$ and $y < 0$. □

## 2

Suppose you have an iterative algorithm for finding a fixed point of a function F(x) on a certain interval. Write a short text (2-4 paragraphs) explaining the following.

1) How do you decide when to stop iterating?

2) What is your strategy for making sure that the computed solution is close to the exact solution?

DO NOT use the standard methodology and estimates from the textbook. Try to find other ways of accomplishing the goals above. Discuss briefly the advantages and disadvantages of your error control approach.

Solution: Halt the algorithm by monitoring the residual function $R(x) = x - F(x)$. The iteration stops when the absolute residual $|R(x_n)|$ drops below a dynamically scaled threshold rather than a static number. This threshold is calculated by multiplying a base tolerance by the local secant slope of the function over the last two iterations. This adaptive stopping criterion allows the algorithm to halt earlier in regions where the function is steep and numerical noise dominates, preventing useless micro-iterations that consume compute time without actually improving the floating-point representation of the answer.

To guarantee the computed solution is close enough to the exact solution we can use a bracketing strategy based on the Intermediate Value Theorem. Once the iteration terminates at a candidate point $x_n$, we evaluate the residual function at the absolute upper and lower bounds of our acceptable error margin by computing $R(x_n - \delta)$ and $R(x_n + \delta)$. If these two evaluations yield opposite signs, and $F(x)$ is known to be continuous, we have absolute mathematical certainty that the true fixed point lies securely within the interval $[x_n - \delta, x_n + \delta]$.

The primary advantage of this error control method is its deterministic, airtight control. It doesn't requiring computing, estimating, or guessing a global Lipshitz constant, which is very difficult to bound analytically for non-trivial functions. It provides a definitive true/false validation of the solution's accuracy without being dependent on the iterative history or path taken to arrive at the candidate point.

The main disadvantage is that the bounding strategy fails if the function does not explicitly cross the line $y = x$. If the derivative at the fixed point is exactly 1—the residual will not change signs across the neighborhood, causing the verification step to yield an inconclusive result even if the computed approximation is perfectly accurate.

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Find the order of convergence for the following:

$$x_n = (1 + 1/n)^{1/2}$$

As $n \to \infty$, the term $1/n \to 0$.

$$x = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{1/2} = \sqrt{1+0} = 1$$

The error at the $n$-th step, $e_n$, is the absolute difference between $x_n$ and the limit $x$:

$$e_n = |x_n - x| = \left(1 + \frac{1}{n}\right)^{1/2} - 1$$

Using the Taylor series expansion for $\sqrt{1+u}$ around $u = 0$, where $u = \frac{1}{n}$:

$$\sqrt{1+u} = 1 + \frac{1}{2}u - \frac{1}{8}u^2 + \mathcal{O}(u^3)$$

Substituting $u = \frac{1}{n}$ back into the series gives:

$$x_n = 1 + \frac{1}{2n} - \frac{1}{8n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)$$

Subtracting the limit $x = 1$ from our expansion leaves the error term:

$$e_n = \frac{1}{2n} - \frac{1}{8n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)$$

For large $n$, the dominant term is $\frac{1}{2n}$. This tells us that the sequence converges to 1 with an asymptotic rate of $\mathcal{O}(1/n)$.

The order of convergence $\alpha$ and the asymptotic error constant $C$ are found using the limit:

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n^\alpha} = C$$

($\alpha = 1$):

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n} = \lim_{n\to\infty} \frac{\frac{1}{2(n+1)}}{\frac{1}{2n}} = \lim_{n\to\infty} \frac{n}{n+1} = 1$$

Because $\alpha = 1$ and $C = 1$, the sequence does not meet the strict definition of linear convergence (which requires $0 < C < 1$). Instead, this sequence exhibits sublinear convergence, specifically converging at a rate of $\mathcal{O}(1/n)$.

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Let $1/2 \leq q \leq 1$, and define $F(x) = 2x - qx^2$. On what interval can it be guaranteed that the method of iteration using F will converge to a fixed point?

$$2x - qx^2 = x$$
$$x - qx^2 = 0$$
$$x(1 - qx) = 0$$

The fixed points are $p_1 = 0$ and $p_2 = \frac{1}{q}$. Since $1/2 \leq q \leq 1$, the target fixed point $p_2$ lies in the interval $[1, 2]$. We analyze convergence to $p = 1/q$.

Let $E_n = \frac{1}{q} - x_n$. Substituting $x_{n+1} = F(x_n)$ yields:

$$E_{n+1} = \frac{1}{q} - (2x_n - qx_n^2) = q\left(\frac{1}{q} - x_n\right)^2 = qE_n^2$$

Multiplying by $q$ gives $qE_{n+1} = (qE_n)^2$, which implies $qE_n = (qE_0)^{2^n}$. Convergence to $1/q$ occurs if and only if $|qE_0| < 1$:

$$|1 - qx_0| < 1 \implies 0 < x_0 < \frac{2}{q}$$

For a specific $q$, the maximal interval of convergence is $(0, \frac{2}{q})$. Because $1/2 \leq q \leq 1$, taking the intersection of these intervals guarantees convergence for any such $q$ on the interval $(0, 2)$.

Contraction condition $|F'(x)| < 1$:

$$-1 < 2 - 2qx < 1 \implies \frac{1}{2q} < x < \frac{3}{2q}$$

Because $F$ maps $(\frac{1}{2q}, \frac{3}{2q})$ into itself, the Contraction Mapping Theorem guarantees local convergence. Taking the intersection of these bounds for $q \in [1/2, 1]$ yields a universal strict contraction interval of $(1, 1.5)$.